

Candidate Name: \_\_\_\_\_ Class: 07S \_\_\_\_\_



# JURONG JUNIOR COLLEGE

## PRELIMINARY EXAMINATION 2008

### H2 BIOLOGY 9747/2

FRIDAY

15 AUGUST 2008

2 HOURS

#### INSTRUCTIONS TO CANDIDATES :

Write your name and class in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working. Do not use staples, paper clips, highlighters, glue or correction fluid.

#### Section A

Answer **all** questions.

Write your answer in the spaces provided on the question paper.

#### Section B

Answer any **one** question.

At the end of the examination

1. fasten any separate answer paper used securely to the question paper.
2. enter the question number from Section B in the grid opposite.

#### INFORMATION FOR CANDIDATES

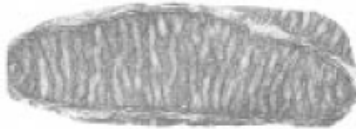
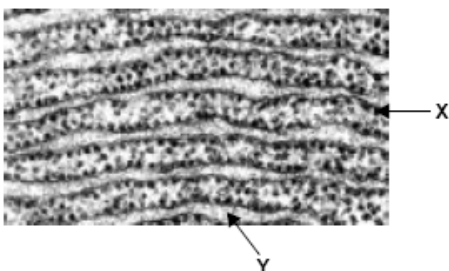
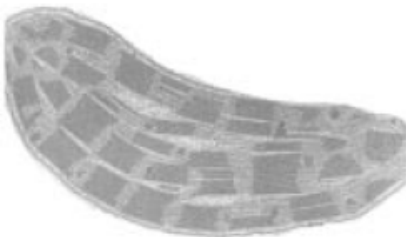
The number of marks is given in brackets [ ] at the end of each question or part question.

| EXAMINER'S USE ONLY |      |
|---------------------|------|
| Section A           | Mark |
| 1                   |      |
| 2                   |      |
| 3                   |      |
| 4                   |      |
| 5                   |      |
| 6                   |      |
| 7                   |      |
| 8                   |      |
| Section B           |      |
|                     |      |
| Total               |      |

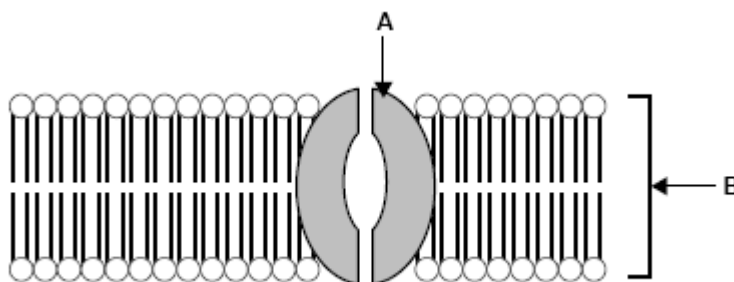
### Section A

Answer **all** the questions in this section.

1. (a) The following table contains images of a number of structures seen in cells. Name each of the structures and outline the function of each. [5]

| Structure                                                                           |      | Name and function                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | i.   | <b>Mitochondrion ;– site of <u>aerobic respiration</u> / production of ATP ;</b>                                                                                                                                     |
|    | ii.  | <b>Structure X</b><br><br><b>Ribosome ;– site of protein synthesis ;</b>                                                                                                                                             |
|                                                                                     | iii. | <b>Structure Y</b><br><b>Rough endoplasmic reticulum ;– site of protein synthesis / <u>protein transport to golgi apparatus</u> ;</b><br><b><i>Accept lumen of endoplasmic reticulum, Reject short form 'ER'</i></b> |
|  | iv.  | <b>Golgi complex / body / apparatus; – <u>modifies, packs and sorts</u> material for transport out of cell ;</b><br><b><i>Reject short form 'GA'</i></b>                                                             |
|  | v.   | <b>Chloroplast ;– site of photosynthesis / production of sugars ;</b>                                                                                                                                                |

The diagram below represents a cross section of part of a cell membrane.



- (b) Name the structures labelled **A** and **B**.

Structure **A - Channel protein** (*reject: 'intrinsic / integral / transmembrane protein'*);;

Structure **B - Phospholipid bilayer** ;; [2]

The concentration of sodium ions, Na<sup>+</sup>, in human blood plasma is approximately 150 mmol/L. In the cytosol of red blood cells the concentration of these ions is between 25 and 30 mmol/L.

(c) Explain how this difference in concentration is maintained. [2]

**Sodium ions are pumped out of the cell against a concentration gradient ;;**  
**This transport is active transport and hence requires energy. ;;**

Animal cells placed in distilled water swell and burst. Plant cells placed in distilled water do not burst.

(d) Explain why plant cells do not burst in this situation. [1]

**A plant cell has a cellulose cell wall outside the cell membrane that limits the expansion of the cell membrane ;;**

[Total: 10]

2. During digestion, large molecules are broken down into their constituent monomers (subunits).

(a) Complete the table below. [2]

| Large molecules                                                                      | Monomers (subunits)               |
|--------------------------------------------------------------------------------------|-----------------------------------|
| <b>Protein ;</b><br><i>Accept named eg.</i>                                          | <b>Amino acids</b>                |
| <b>Lipid</b>                                                                         | <b>Fatty acids and glycerol ;</b> |
| <b>Polysaccharide ;</b><br><i>Reject: 'carbohydrates'</i><br><i>Accept named eg.</i> | <b>Monosaccharide</b>             |
| <b>DNA</b>                                                                           | <b>Deoxyribonucleotides ;</b>     |

*Students were awarded two marks for four correct answers; minus one mark for every mistake made.*

(b) Explain how a monosaccharide enters an epithelial cell. [2]

- **A monosaccharide moves by facilitated diffusion or active transport;;**
- **This occurs via channel proteins or by carrier proteins molecules;;**

***Reject "endocytosis" → bulk transport***

(c) For what purpose would a cell use a lipid monomer? [1]

**Either one of:**

- **as a component of plasma membrane;;**
- **as an energy source;;**

Starch, a storage molecule in plants, is a polymer made of glucose molecules joined together. It consists of two components.

(d) Name these two components. [2]

- **amylose;;**
- **amylopectin;;**

(e)(i) Give three similarities between these two components. [3]

- contain  $\alpha$ -glucose monomers;;
- coiled into a helix;;
- no cross-linking;;
- contain  $\alpha(1\rightarrow4)$  glycosidic bonds;;

(ii) Give two differences between these two components. [2]

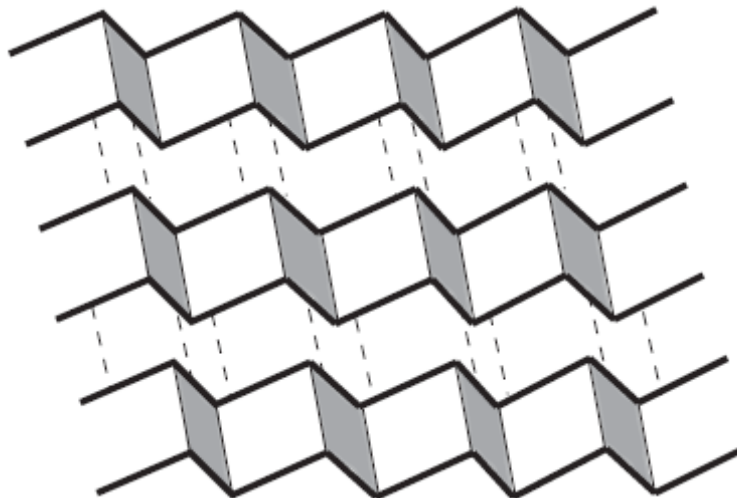
- The amylose component is unbranched whilst the amylopectin component is branched;;
- The glucose residues in the amylose component are linked by  $\alpha(1\rightarrow4)$  glycosidic bonds, while the glucose residues in the amylopectin component are linked by both  $\alpha(1\rightarrow4)$  glycosidic bonds and  $\alpha(1\rightarrow6)$  glycosidic bonds;;

Scientists are now turning to the study of the proteome (all of the proteins) of an organism rather than the study of single proteins.

(f) Suggest **one** reason why the emphasis is now on the study of all the proteins of an organism rather than on one protein at a time. [1]

**The emphasis is on the study of all proteins because of the interaction between proteins to carry out most functions;;**

Protein molecules come in many shapes and forms that can be classified into primary, secondary, tertiary and quaternary. The secondary, tertiary and quaternary shapes arise as a result of different kinds of folding of a primary structure. One kind of secondary structure is a pleated sheet where the primary molecule extends along the folded sheet. The primary structures in the layers are held together by hydrogen bonding.



(g) Suggest two ways in which such a structure may be important to the functioning of the protein. [2]

**Structures may be important because:**

- of the ability of the protein to stretch or contract / elongate or shorten / be flexible in particular situations;;
- pleating may strengthen the molecule / provide tensile strength and that may be important for its function;;
- particular structures may provide an active or binding site for an enzyme or other molecule;;

[Total: 15]

3. In *Drosophila* flies, a gene for wing shape found on chromosome 2 has two alleles – allele **A** for curly wing, which is dominant to allele **a** for normal wing. Female flies are homogametic while males are heterogametic. The gene for *Drosophila* eye colour is located on the X chromosome and has two alleles – the allele for red eyes being dominant to the allele for white eyes.

(a)(i) Write the genotypes of the two flies in the table below. [2]

|            | Fly 1                    | Fly 2                    |
|------------|--------------------------|--------------------------|
| Sex of fly | Female                   | Male                     |
| Eye colour | White                    | Red                      |
| Wing shape | Curly wing, heterozygous | Curly wing, heterozygous |
| Genotype   | $X^r X^r Aa$             | $X^R Y Aa$               |

**1 mark for each correct genotype**

- (ii) Draw a genetic diagram showing the phenotypes of the offspring and the expected ratio of phenotypes when these two flies are crossed. [4]

**Parental genotype:**  $X^r X^r Aa$  x  $X^R Y Aa$

**Gametes :**  $(X^r A)$   $(X^r a)$  x  $(X^R A)$   $(X^R a)$   $(YA)$   $(Ya)$

**Offspring Genotypes:**  $X^R X^r AA$   $X^R X^r aa$   $X^r Y AA$   $X^r Y aa$   
 2  $X^R X^r Aa$  2  $X^r Y Aa$

**Offspring Phenotypes:** Red eyed Curly wing Female Red eyed Normal wing Female White eyed Curly wing Male White eyed Normal wing Male

**Offspring phenotypic ratio:** 3 : 1 : 3 : 1

240 offspring were obtained from the above cross, and the phenotypes of the offspring were as follows:

- 83 red-eyed females with curly wings
- 41 red-eyed females with normal wings
- 78 white-eyed males with curly wings
- 38 white-eyed males with normal wings

| Degrees of freedom | Probability, p |      |       |       |       |
|--------------------|----------------|------|-------|-------|-------|
|                    | 0.10           | 0.05 | 0.02  | 0.01  | 0.001 |
| 1                  | 2.71           | 3.84 | 5.41  | 6.64  | 10.83 |
| 2                  | 4.61           | 5.99 | 7.82  | 9.21  | 13.82 |
| 3                  | 6.25           | 7.82 | 9.84  | 11.35 | 16.27 |
| 4                  | 7.78           | 9.49 | 11.67 | 13.28 | 18.47 |

$\chi^2$  test:  $\chi^2 = \sum \frac{(O - E)^2}{E}$   $v = c - 1$

Key to symbols:  $\Sigma$  = sum of ...  
 $v$  = degrees of freedom  
 $c$  = number of classes  
 $O$  = observed value  
 $E$  = expected value

Using the formula for the  $\chi^2$  test and the tables of  $\chi^2$  values above,

(b)(i) Calculate the value of the  $\chi^2$  test on these results. Show your working in the space provided below. [2]

$$\chi^2 = \frac{(83-90)^2}{90} + \frac{(41-30)^2}{30} + \frac{(78-90)^2}{90} + \frac{(38-30)^2}{30} = 8.31$$

**1 mark for working;;**

**1 mark for final answer;;**

(ii) State the degree of freedom for these results. [1]

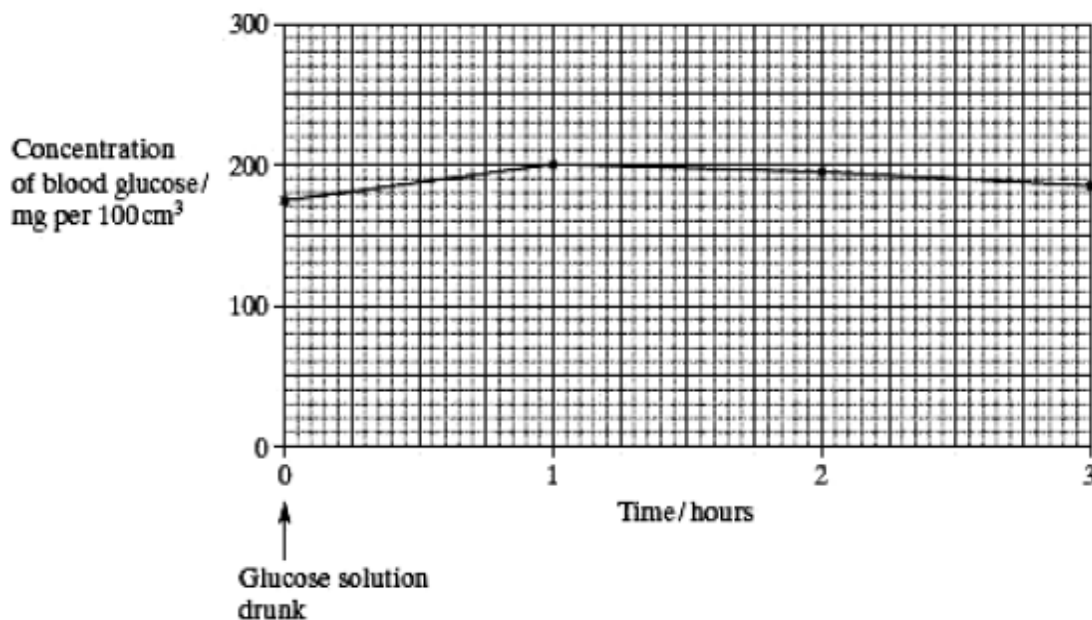
**3 ;;**

(iii) Does this value for the chi-squared test support the expected phenotypic ratio that you have obtained in (a)(ii)? Explain. [3]

- **No;;**
- **Calculated chi square value of 8.31 is greater than the critical value of 7.82;**
- **at  $p = 0.05$ ;**
- **Hence difference between observed and expected results is significant;**
- **and is not due to chance;**

[Total: 12]

4. Some people produce no insulin. As a result, they suffer from a condition known as diabetes. During an investigation, a man with diabetes drank a glucose solution. The concentration of glucose in his blood was measured at regular intervals. The results are shown in the graph below.



(a) Suggest one reason why the concentration of glucose decreased after 1 hour even though this man's blood contained no insulin. [1]

- **glucose is used in cell respiration / as energy source / in metabolism (*must qualify how glucose is used*);;**
- **glucose is excreted / removed in urine;;**

- (b) The investigation was repeated on a man who did not suffer from diabetes. The concentration of glucose on this man's blood before drinking the glucose solution was  $80\text{mg per } 100\text{cm}^3$ . Sketch a curve on the graph above to show the results you would expect. [1]

**line starting from 80mg, increasing but keeping below line for diabetic, dropping to 80mg (line must stabilise at, or fluctuate around 80mg), to complete graph;;**

- (c) The diabetic man adopted a daily routine to stabilize his blood glucose concentration within narrow limits. He ate three meals a day: breakfast, a midday meal and an evening meal. He injected insulin once before breakfast and once before the evening meal.

The injection he used before breakfast was a mixture of two types of insulin - slow-acting insulin and fast-acting insulin.

- (i) Explain the advantage of injecting both types of insulin before breakfast. [2]

- fast acting insulin reduces blood glucose from breakfast ;;**
- slow acting insulin reduces blood glucose from lunch / eliminates the need to inject at lunch;;**  
**(must make reference to the meals)**

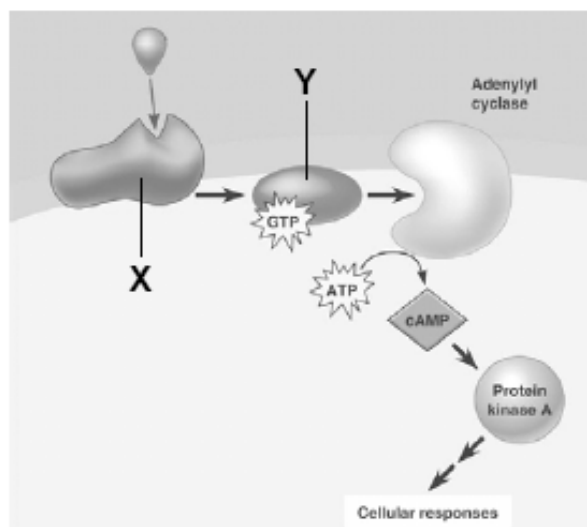
- (ii) One day, the man did not eat a midday meal. Suggest one reason why his blood glucose concentration did not fall dangerously low even though he had injected himself with the mixture of insulin before breakfast. [1]

- glucagon is still produced to convert glycogen to glucose / to stimulate glycogenolysis;;**
- slow-acting version present will not reduce blood glucose drastically;;**
- person is not active so little glucose used in respiration;;**  
**(any one, REJECT statements about consuming large breakfasts)**

[Total: 5]

5. Certain microbes cause disease by disrupting the G-protein signaling pathways. The cholera bacterium, *Vibrio cholerae*, present in water contaminated with human waste colonizes the small intestine and produces a toxin that **modifies a G protein**.

The toxin results in a high concentration of cAMP which causes the intestinal cells to secrete large amounts of water and salt into the intestines, leading to profuse diarrhea and death from the loss of water and salts.



(a) Name the molecules **X** and **Y**. [2]

- **X = G-protein linked receptor;; (reject: receptor protein / transmembrane protein)**
- **Y = G-protein;;**

(b) What is the general name given to small molecules that function like cAMP in signaling pathways? [1]

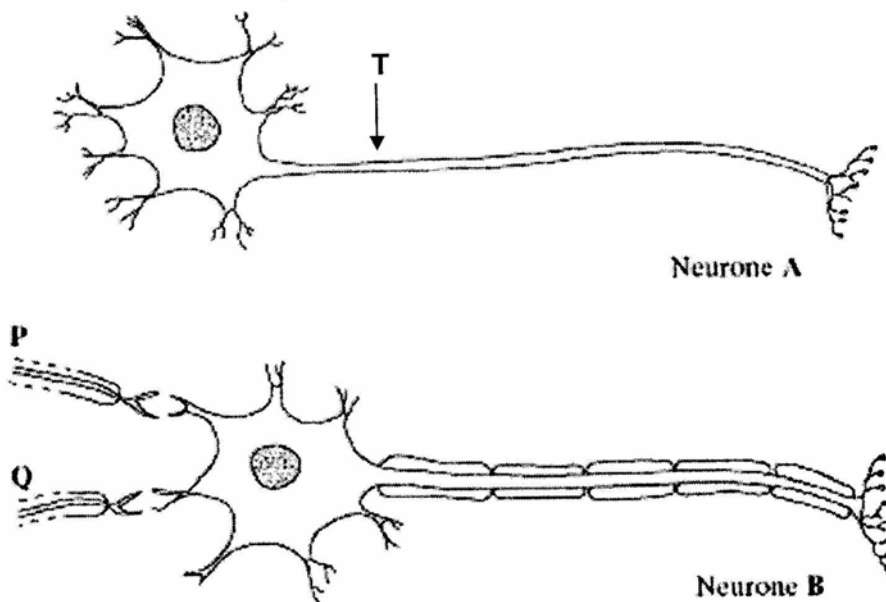
- **Secondary / second messengers;;**

(c) Suggest how the toxin leads to an increase in cAMP concentration. [3]

- **modified G protein binds to GTP permanently / unable to hydrolyze / break down GTP to GDP;;**
- **remains in its active form;;**
- **adenylyl cyclase is continuously stimulated to make cAMP;;**

[Total: 6]

6. The figure below shows two motor neurons, **A** and **B**. It also shows the synapses of neurone **B** with two other neurones, **P** and **Q**.



(a)(i) If an action potential is initiated at point **T** on neurone **A**, describe how this action potential is propagated **along** the neurone. [3]

- **When the neurone is stimulated, Na<sup>+</sup> rush into the axon along a concentration gradient. This causes depolarization of the membrane;;**
- **Propagation of impulses due to localised electrical circuits as a result of diffusion of Na<sup>+</sup> to adjacent region;;**
- **Opening of Na<sup>+</sup> gated channels at the adjacent region results in influx of Na<sup>+</sup> ions and subsequent progression of the impulse;;**
- **Behind the impulse, K<sup>+</sup> begin to leave the axon along a concentration gradient, repolarization of membrane;;**

(After the impulse has passed and the neurone is repolarised, Na<sup>+</sup> is once again actively expelled in order to increase the external concentration and so allow the passage of another impulse)

[max 3 marks]



(ii) Suggest why transmission of impulses along neurone **B** uses less energy than transmission along neurone **A**. [2]

- Neurone B is myelinated so saltatory conduction occurs / where impulses / AP jump from node to node;;
- Active transport of ions occurs at the nodes of ranvier only / less sodium-potassium pumps are required, hence less respiration / ATP required to maintain resting potential;;

(b) GABA is a neurotransmitter present in some parts of the nervous system. The diagram below shows how the release of GABA from a presynaptic membrane affects the ion channels of a postsynaptic membrane.



(i) Explain what would happen to the membrane potential on the postsynaptic membrane when GABA is released. [2]

- Membrane potential becomes more negative / decreases below resting potential / hyperpolarized;;
- GABA causes opening of potassium ion channels, resulting in more potassium ions diffusing out of the postsynaptic membrane;;
- GABA causes opening of chloride ions channels, resulting in more chloride ions diffusing into the postsynaptic membrane;;

(ii) Name the process whereby neurone **B** receives and integrates signals from neurones **P** and **Q** at the same time. [1]

- Spatial summation;;

(iii) Explain why neurone **B** is less likely to respond if both neurones **P** and **Q** are stimulated at the same time but neurone **P** releases acetylcholine whereas neurone **Q** releases GABA. [2]

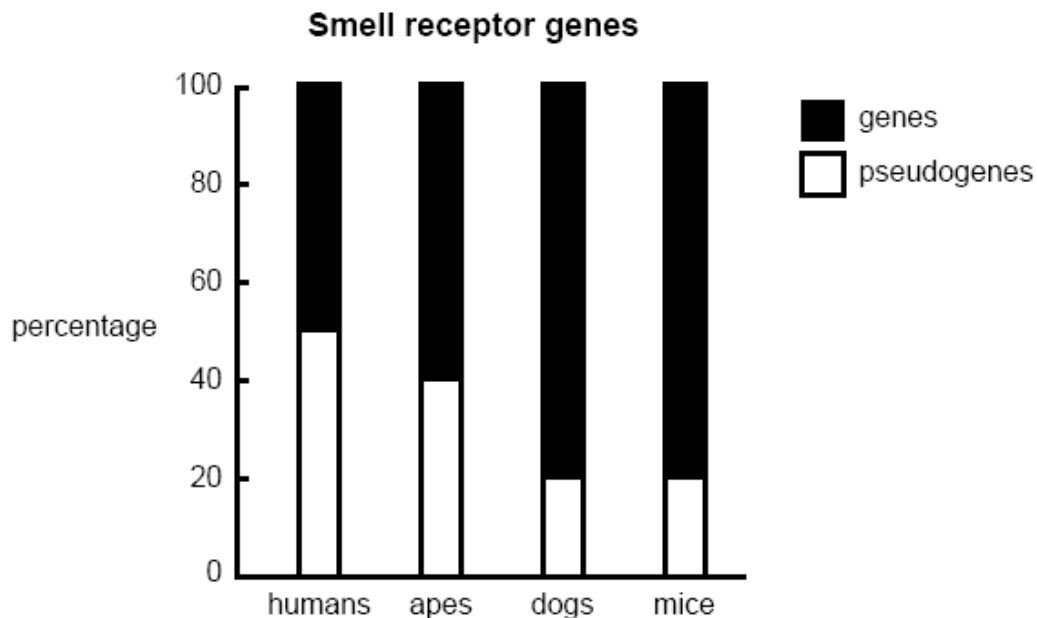
- Acetylcholine from P will generate an excitatory postsynaptic potential while GABA from Q will generate an inhibitory postsynaptic potential;; (Reject: "EPSP/IPSP")
- The summation of these potentials cancels / counteracts each other, resulting in a weak stimulus in neurone B which does not reach threshold.

Total: 10 marks, each "•" = 1 mark unless otherwise stated.

[Total: 10]

7. (a) Pseudogenes are the remains of broken genes which are unable to function and can be considered to be genetic fossils. Some are relics of genes lost through evolution while others reflect an earlier version of a present functional gene. Pseudogenes are able to accumulate all kinds of mutations.

In mammals, more than one thousand genes coding for smell receptors have been identified. Individual species vary in the proportion of these genes that have become pseudogenes, and humans have fewer than 500 functional genes coding for smell receptors. The majority of these genes are still functional genes in rats and mice.



- (i) From the [graph](#), what can be [concluded](#) about the importance of smell to the survival of these [four groups](#)? [2]

- **Smell is most important to dogs and mice;;**
- **less important to apes and least important to humans;;**

**NOTE: Students are required to use the names of the specific examples given rather than make general statements.**

- (ii) Mice and dogs can only distinguish shades of grey, while some apes and monkeys are able to distinguish colours. [Would you expect to find any change](#) in the numbers of smell pseudogenes in apes and monkeys able to distinguish colours? Explain why. [2]

- **Yes;;**
- **Apes and monkeys are less dependent on smell / more dependent on vision for survival;**
- **They would therefore have more pseudogenes / less smell genes;**

**NOTE: Students are to make the link that, as smell is less important to these animals, mutations are possible in genes responsible for smell as these will not affect the organism's chance of survival. These mutated genes are classed as pseudogenes.**

**Senses/systems work together to ensure survival, even though genes involved may be on separate gene loci!**

- (iii) Why are pseudogenes able to accumulate mutations that do not exist in functional genes? [2]

- **Pseudogenes are not expressed / non-coding and any mutation in them does not harm the offspring when passed on;;**

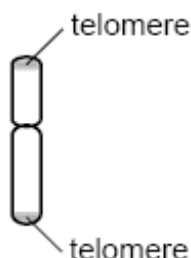
- **Functional genes become disrupted / produce a different protein which could affect the survival of offspring, thus less likely to be passed on (*idea to be conveyed at least once*) to subsequent generations;;**

(b) Body cells have elaborate control systems that regulate their division. If these control systems break down, cells start to grow and divide in an unregulated way and produce a tumour.

(i) Cigarette smoking can lead to the formation of tumours. Describe how. [3]

- **Contains carcinogens/cancer-causing substances;**
- **Causes mutation/changes in DNA/genes/genetic material;**
- **Of genes controlling normal rate of cell division/regulating cell cycle;**
- **Tumour suppressor genes becomes less active / loss-of-function; (reject: tumour suppressing gene)**
- **Proto-oncogenes become oncogenes / more active / gain-of-function; (if only 'tumour suppressor genes' and 'proto-oncogenes' stated without mention of activity levels → 0.5 marks)**
- **Leads to uncontrolled cell division/mitosis;**

Chromosomes are DNA structures in the nucleus of a cell. The ends of chromosomes are called telomeres.



Some changes in the DNA of these telomeres can trigger the development of cancer. Scientists are working to detect these changes in telomeres at an early stage of their development.

(ii) Outline **one advantage of being able to detect early changes** in telomere DNA. [1]

**The early detection of changes in the DNA could lead to:**

- **the cancer being treated earlier;;**
- **members of the family receiving advice to be tested;;**
- **a lifestyle change to reduce further risk;;**

**NOTE: Reject if students stated the cancer could be detected but made no mention of why this was an advantage. The knowledge gained by the change in the DNA has to be used to advantage.**

One normal sequence of DNA in a particular telomere was AATCGTCCAGGG. On examination of the telomeres of one patient, this sequence was found to have changed to AACCGTCCTGGG. The doctor examined the relationship between a DNA sequence and amino acids that could be expected. He used the following table to obtain information.

| DNA sequence | Amino acid |
|--------------|------------|
| AAT or AAC   | leucine    |
| CGT or CGG   | alanine    |
| CCA or CCT   | glycine    |
| GGG or GGT   | proline    |

(iii) Is the doctor likely to be concerned or not concerned about the change in the telomere DNA of this patient? Explain your answer. [2]

- The doctor is not likely to be concerned;;
- as the changes in the DNA produces the same amino acid sequence / same aa incorporated into protein;;
- since the genetic code is degenerate i.e. more than 1 codon may code for the same aa;; (reject: DNA is degenerate / amino acid is degenerate)

[Total: 12]

8. Cytochrome c is found in the mitochondria of every aerobic eukaryote — animal, plant, and fungi. The amino acid sequences of cytochrome c from many of these organisms have been determined, and comparison made between the molecules shows that they are related.

Human cytochrome c contains 104 amino acids, and 37 of these have been found at equivalent positions in every cytochrome c that has been sequenced. **These molecules are homologous.**

Table 3 shows the first 18 amino acid residues of human cytochrome c, from the N-terminal, with the corresponding sequences from six other organisms aligned beneath. A ditto (") indicates that the amino acid is the same one found at that position in the human molecule. All the vertebrate cytochromes (the first four) start with glycine (Gly). The *Drosophila*, wheat, and yeast cytochromes have several amino acids that precede the sequence shown here (indicated by <<<).

| Cytochrome        | Amino acid |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-------------------|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|                   |            | 1   |     |     |     |     | 6   |     |     |     | 10  |     |     |     | 14  |     |     | 17  | 18  |
| Human             |            | Gly | Asp | Val | Glu | Lys | Gly | Lys | Lys | Ile | Phe | Ile | Met | Lys | Cys | Ser | Gln | Cys | His |
| Pig               |            | "   | "   | "   | "   | "   | "   | "   | "   | "   | "   | Val | Gln | "   | "   | Ala | "   | "   | "   |
| Chicken           |            | "   | "   | Ile | "   | "   | "   | "   | "   | "   | "   | Val | Gln | "   | "   | "   | "   | "   | "   |
| Dogfish           |            | "   | "   | "   | "   | "   | "   | "   | "   | Val | "   | Val | Gln | "   | "   | Ala | "   | "   | "   |
| <i>Drosophila</i> | <<<        | "   | "   | "   | "   | "   | "   | "   | "   | Leu | "   | Val | Gln | Arg | "   | Ala | "   | "   | "   |
| Wheat             | <<<        | "   | Asn | Pro | Asp | Ala | "   | Ala | "   | "   | "   | Lys | Thr | "   | "   | Ala | "   | "   | "   |
| Yeast             | <<<        | "   | Ser | Ala | Lys | "   | "   | Ala | Thr | Leu | "   | Lys | Thr | Arg | "   | "   | "   | "   | "   |

Table 3

(a) With reference to Table 3, explain what is meant by "These molecules are homologous" (line 6-7). [3]

- The cytochrome molecule of the various organisms is a shared characteristic derived from a common ancestor ;;
- The cytochrome molecules share the same amino acids at many positions ;;
- For all organisms, the molecules have the same amino acid at positions 1,6,10,14, 16, 17 and 18, from the N-terminal ;;

(b)(i) Explain why the analysis of amino acid sequence is used to establish evolutionary relationship. [3]

- The amino acid sequence of a protein is defined by the DNA sequence;;
- Mutations to the cytochrome c gene became fixed in a population / part of the genome and were inherited ;;
- Over time, as each population accumulates its own set of mutations, the original DNA sequence diverges and is reflected in the amino acid sequence. ;;
- The more similar the amino acid sequence, the more recently the divergence took place / more closely related the two organisms are ;;

[max 3]

(ii) State a limitation of using difference in amino acid sequences of only one protein (cytochrome c) to establish phylogenetic relationship among the organisms. [1]

- **One protein represents only a very small percentage of the genetic information in the organisms;;**
- **Is not representative of the entire genome;;**

Cytochrome c is an ancient molecule which has coding and non-coding regions. It has evolved very slowly. Even after more than 2 billion years, one-third of its amino acids are unchanged. The conserved sequences are a great help in working out the evolutionary relationships between distantly-related organisms like fish and humans. However, for humans and the apes (e.g. chimpanzee), their cytochrome c molecules are identical and evolutionary relationships cannot be determined.

(c) State the name given to  
Coding regions – **exons;;**      Non-coding regions – **introns;;** [2]

(d) Suggest an alternative way by which the evolutionary relationship between humans and the apes can be determined by molecular techniques. [1]

- **Comparing DNA sequences of other proteins;; OR**
- **Compare mRNA sequences of other proteins;; OR**
- **Compare amino acid sequence of other proteins;;**

[Total: 10]

Write your answers on the separate answer paper provided.  
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.  
Your answers must be in continuous prose, where appropriate.  
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

9. (a) Describe the structure of DNA and its arrangement in a chromosome. [8]

1. **Double helix;;**
2. **2 anti - parallel strands with 3' and 5' ends;;**
3. **Specific nucleotide sequence along each DNA strand;;**
4. **Each nucleotide consists – deoxyribose sugar, phosphoric acid and nitrogenous base;;**
5. **4 bases - Adenine, Thymine, Cytosine and Guanine (spell out for the first time);;**
6. **Nitrogenous bases projecting at right angle within the helix;;**
7. **0.34nm between adjacent base pairs;;**
8. **Hydrogen bonds between base pairs;;**
9. **Complementary base-pairing: A with T, C with G;;**
10. **Sugar-phosphate backbone with phosphodiester links (bonds);;**
11. **Width between the two sugar-phosphate backbone is 0.2nm;;**
12. **10 bases pairs in one complete turn of helix / 3.4nm per turn;;**

[Total marks = 12, max 4 marks]

13. **DNA wound around histone octamer/histones to form nucleosome;;**
14. **Formation of 'beads on a string';;**
15. **Strings of nucleosomes coiled to form a 30nm chromatin fiber / solenoid;;**
16. **Looped domains of the 30nm fiber / 30nm fiber forms loops;;**
17. **Attached to a scaffold of non-histone proteins;;**
18. **Further looping and coiling to form metaphase chromosome;;**

[Total marks = 6, max 4 marks]

[Total marks = 18, max 8 marks]

(b) Compare and contrast between DNA replication and transcription. [12]

**Similarities:**

1. **Both involve unwinding of the double helix DNA;;**
2. **Both involve breaking of the weak hydrogen bonds between complementary bases;;**
3. **Both involve formation of phosphodiester bonds between adjacent nucleotides;;**
4. **Both involve aligning free nucleotides through complementary base pairing;;**
5. **Both products elongate in 5' to 3' direction;;**
6. **Both occur in the nucleus;;**

[Total marks = 6, max 5 marks]

**Differences:**

| Features                                                | DNA replication                                                                                                                                                                                                 | Transcription                                                                                             |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1. (Main) Enzyme involved                               | <u>DNA polymerase</u> - binds to the parental DNA molecule at the origin of replication;                                                                                                                        | <u>RNA polymerase</u> - binds to the promoter, a region near the beginning of the gene to be transcribed; |
| 2. Raw Material                                         | Deoxyribonucleotides;                                                                                                                                                                                           | Ribonucleotides;                                                                                          |
| 3. Template                                             | Both strands of DNA molecule act as a template;                                                                                                                                                                 | One strand of DNA molecule acts as the template;                                                          |
| 4. Base Pairing                                         | Adenine with Thymine and Cytosine with Guanine;                                                                                                                                                                 | Adenine with Uracil and Cytosine with Guanine;                                                            |
| 5. Presence of proofreading property on enzyme involved | DNA polymerase proofreads the newly synthesized daughter strand, ensuring precise complementary base pairing;                                                                                                   | No proof-reading function;                                                                                |
| 6. Products                                             | 2 DNA molecules;                                                                                                                                                                                                | 1 RNA molecule;                                                                                           |
| 7. Location of the products                             | Products remain in the nucleus;                                                                                                                                                                                 | Products leave the nucleus;                                                                               |
| 8. Post – transcriptional modification                  | Products do not go through post - transcriptional modification;                                                                                                                                                 | Products / pre-mRNA may go through post-transcription modification / splicing / capping / poly-A tailing; |
| 9. When the process occur                               | Prior to nuclear division / mitosis / meiosis;                                                                                                                                                                  | Prior to translation;                                                                                     |
| 10. Purpose                                             | Double the amount of DNA so that after mitosis, the daughter cells will have the same amount of DNA as the parent cell / after meiosis, the daughter cells will have half the amount of DNA as the parent cell; | For subsequent protein synthesis;                                                                         |

[Total marks = 10, max 7 marks]

*If no point-by-point comparison for 'differences', to half total no. of marks obtained.*

10. (a) Explain the importance of the light dependent reaction in photosynthesis. [7]

1. Photosynthesis consists of light dependent reaction and light independent reaction;;
2. Purpose of photosynthesis: harvesting light energy in light dependent reaction to produce carbohydrates in light independent reaction;;
3. Light reaction consists of non-cyclic photophosphorylation and cyclic photophosphorylation;;

#### Non-cyclic photophosphorylation

4. Produces ATP and NADPH in roughly equal amounts;;
5. Light energy is absorbed by light harvesting centres of PSII and PSI which energizes electrons in special chlorophyll a in the reaction centres which is boosted to a higher energy level;;
6. Electrons are passed down an ETC and energy released is used to pump protons from stroma into thylakoid space;;
7. This creates a proton gradient / proton motive force generated;;
8. When  $H^+$  moves down a concentration gradient via ATP synthases / stalked particles, energy lost is used to phosphorylate ADP to ATP (WTTE);;
9. Electrons from PSI is eventually transferred via 2<sup>nd</sup> ETC to  $NADP^+$  to produce NADPH;;

#### Cyclic photophosphorylation

10. High energy electrons from PSI returns back to its starting point via an ETC;;
11. To generate more proton motive force for ATP production;;

#### Uses in light independent reaction / Calvin cycle

12. ATP provides energy;;
13. and NADPH provides reducing power;;

[Total marks = 13, max 7 marks]

(b) Describe the role of  $NAD^+$  in aerobic respiration. [6]

1. co-enzyme;;
2.  $H^+$  and electron carriers;;
3. carry protons and electrons to the ETC for oxidative phosphorylation when reduced to NADH;;
4. Role of  $NAD^+$  in glycolysis:  
Two molecules of Triose phosphate (3C) are phosphorylated and oxidised to form two molecules of Glycerate biphosphate (3C)/ 1,3-diphosphoglycerate;;



**5. Role of NAD<sup>+</sup> in Link reaction:**

Two pyruvate molecules (3C) undergo oxidation decarboxylation to form two molecules of acetyl CoA (2C);;

**6. Role of NAD<sup>+</sup> in Krebs cycle:**

Citrate (6C) undergoes oxidative decarboxylation to form  $\alpha$ -ketoglutarate (5C) /  
 $\alpha$ -ketoglutarate (5C) undergoes oxidative decarboxylation to form succinate (4C) /  
Succinate (4C) undergoes oxidation to form fumarate (4C);;

**7. Role of oxidative phosphorylation:**

For the synthesis of 30 out of the 38 ATP produced during the complete oxidation of 1 glucose molecule from 10 molecules of NADH produced in glycolysis, Link reaction and Krebs cycle;;

[Total marks = 7, max 6 marks]

(c) Compare and contrast photophosphorylation and oxidative phosphorylation. [7]

**Similarities:**

1. involve transfer of electrons between electron carriers
2. energy released from electron transport is used to generate proton gradient
3. potential energy of proton gradient is harnessed for ATP synthesis

**Differences:**

| Features                              | Photophosphorylation                                            | Oxidative Phosphorylation                                          |
|---------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| 1. Sources of energy                  | energy for making ATP comes from light                          | energy for making ATP comes from oxidation processes               |
| 2. Location                           | Thylakoid membrane of chloroplast                               | Inner membrane of mitochondria                                     |
| 3. Involvement of light energy        | Required to energize the electrons in special chlorophyll a     | Not required                                                       |
| 4. Source of energy for ATP synthesis | Light                                                           | Glucose oxidation                                                  |
| 5. Electron donors                    | For non-cyclic reaction: water;<br>For cyclic reaction: PS I    | NADH, FADH <sub>2</sub>                                            |
| 6. Electron acceptors                 | Non-cyclic acceptor: NADP <sup>+</sup><br>Cyclic acceptor: PS I | Oxygen                                                             |
| 7. Establishment of proton gradient   | H <sup>+</sup> pumped inwards, from stroma to thylakoid space   | H <sup>+</sup> pumped outwards, from matrix to intermembrane space |

[Total marks = 10, max 7 marks]